Hankuk University
CSE 204 Data Structures
Credits: 3

Course Description
Fundamental data structures and their algorithms, with analysis of running time.

Learning Outcomes
1. Analyze algorithms with asymptotic notation.
2. Implement linked lists, stacks and queues.
3. Implement binary search trees, heaps and hash tables.
4. Implement and compare sorting algorithms and basic graph traversal.

Textbook
Goodrich, Tamassia and Mount, Data Structures and Algorithms in C++

Prerequisites
CSE 103 C Programming

Weekly Schedule
Week 1: Algorithm analysis and Big-O notation
Week 2: Arrays and linked lists
Week 3: Stacks and queues
Week 4: Recursion
Week 5: Trees and binary trees
Week 6: Binary search trees
Week 7: Balanced search trees
Week 8: Midterm exam
Week 9: Priority queues and heaps
Week 10: Hash tables
Week 11: Sorting: merge sort and quick sort
Week 12: Graphs: representation, breadth-first and depth-first search
Week 13: Shortest paths
Week 14: Review
Week 15: Final exam

Assessment
Assignments 30%, Midterm 30%, Final exam 40%